NUST Institute of Civil Engineering
School of Civil & Environmental Engineering

CS-114: Fundamentals of Programming

BECE-2018 (Fall 2018)

One Hour Test (OHT) - II

28th Nov, 2018

Total Marks: 30 **Total Time Allowed: 60 mins**

Name: _____ Reg#: _____

Course: _____ Section: _____

Instructions:

1. This is a closed-book, closed-notes examination.
2. This exam consists of two sections.
3. Write your name and registration number on this page.
4. **Solve the questions in the space provided. Do not use answer sheets. Use backside of the paper to do rough work.**
5. Mobile phones are not allowed in the examination room. Possession of it will result in immediate disqualification.

CLO : BTL	CLO 3 : C4	CLO 2 : C3
Section.Question	II	III
Total Marks: 30	15	15
Marks Obtained:		
CLO: Course Learning Outcome, BTL: Bloom's Taxonomy Level *C = Cognitive domain, P = Psychomotor domain, A = Affective domain*		

_____ _____

Instructor: Lec. Qurrat-ul-ain Babar **Reviewed by:**

_____

Invigilator:

SECTION – I [15 Marks]

1. ***[5 marks]*** **Analyse** the following code snippets to find the **output** after C++ statements are executed?

a)
```
int g = 15;
int h = 8;

int main () {
   int g = 8;
   cout << g << h;
}
```

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b)
```
int n = 3;
while (n >= 0)
{
    cout << n * n;
    --n;
}
cout << endl << n << endl;

while (n < 4)
     cout << ++n;
cout << endl << n << endl;

while (n >= 0)
     cout << (n /= 2) << endl;
```

9410
-1
01234
4
2
1
0
0
...

2. ***[5 marks]*****Analyze** the questions below concerning the following fragment of code to give their **answers**.

```
int n;
cout << "Enter an integer: ";
cin >> n;

if (n < 10)
  cout << "less than 10" << endl;
else if (n > 5)
  cout << "greater than 5" << endl;
else
  cout << "not interesting" << endl;
```

1. What will be the output of the fragment above if the user enters the integer value 0 ?

____less than 10_____

2. What will be the output of the fragment above if the user enters the integer value 15 ?

_____greater than 5_____

3. What will be the output of the fragment above if the user enters the integer value 7 ?

_____less than 10_____

4. What values for n will cause the output of the fragment above to be "`not interesting`"?

_____no values_____

3. [5 marks] Given below is one way of calculating 1+2+...+n:

```
// Pre-condition: n is 1 or greater
// Post-condition: this functions returns the value of
// 1+2+3+...+n
// Examples: sum(1) is 1, sum(2) is 3, sum(3) is 6, ...
int sum(int n)
{
   int sum = 0;

   while (n > 0) {
     sum = sum + n;
     --n;
   }
   return sum;
}
```

Rewrite the body of the ***sum*** function using a single for-loop instead of a while-loop.

```
int sum(int n)
{
  int sum = 0;

  for (; n>0; n--)
  {
   sum += n;
  }
  return sum;
}
```

SECTION – II [15 Marks]

Develop programs in C++ for following problems:

1. ***[4 marks]*** The factorial function n! is defined by

   $$n! = n.n\text{-}1.n\text{-}2....3.2.1$$

   and 0! Is defined to be 1. **Develop** a C++ **function** called "`fact`" whose input is a non-negative integer `n` and which returns the value of n!.

```
int fact(int n)
{
   int factorial = 1;

   if(n <= 1)
   {
     factorial = 1;
   }
   else
   {
     for (; n>0; n--)
     {
          factorial *= n;
     }
   }
   return factorial;
}
```

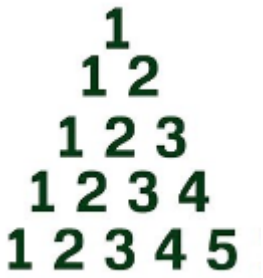
2. ***[5 marks]*** **Develop** a **function** named "`enough`" that takes one integer argument, call it "`goal`" and returns as its value the smallest *positive* integer n for which 1+2+3+. . . +n is at least equal to goal. Thus, for example,

```
cout << enough(9) << endl; // will print 4 because 1+2+3+4>9 but 1+2+3<9
cout << enough(21)<< endl; //  will print 6 'cause 1+2+ . . .+6 >21 but 1+2+ . . . 5<21
cout << enough(-7)<< endl; // will print 1 because 1 > -7 and 1 is the smallest
                           // positive integer
cout << enough(1) << endl; // will print 1 because 1 = 1 and 1 is the smallest
                           // positive integer
```

```
int enough(int goal)
{
   int n = 1;
   int sum = 0;

   if(goal <= 1)
   {
     return n;
   }
   else
   {
     while(sum <= goal)
     {
          sum = sum + n;
          ++n;
     }
   }
   return n;
}
```

3. ***[6 marks]*** **Develop** a **program** which takes number of rows as input from the user and prints the following pattern on screen. For example, if a user enters 5 as number of rows, the output will be:




```
#include <iostream>
using namespace std;

int main()
{
   int n;
   cout << "Enter number of rows:";
   cin >> n;

   for (int i=1; i<=n; i++)
   {
     for(int j=1; j<=n-i; j++)
          cout << " ";

     for(int k=1; k<=i; k++)
          cout << k <<" ";

     cout << endl;
   }
   return 0;
}
```

THE END